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# Khulna University, Khulna

## Mathematics Discipline

4<sup>th</sup> Year, Term-I Examination; Session: 2019-2020

Course No: Math-4107 (Old); Title of the Course: Differential Geometry

Full Marks: 60, Time: 03 Hours

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

### Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                                                                 | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. (a) Define parametric equations. What is well behaved and badly behaved curves?                                                                              | 04    |
| (b) Draw the graph of the vector valued functions:                                                                                                              | 06    |
| (i) $r(t) = \cos t \hat{i} + \sin t \hat{j}; 0 \leq t \leq 2\pi$                                                                                                |       |
| (ii) $r(t) = \cos t \hat{i} + \sin t \hat{j} + 2t \hat{k}; 0 \leq t \leq 2\pi$ .                                                                                |       |
| 2. (a) Define the derivative of the vector-valued function. Explain why $r'(t)$ is always directed to the positive side.                                        | 05    |
| (b) Find the parametric equations of the tangent line to the circular helix $x = \cos t, y = \sin t, z = t$ .                                                   | 05    |
| 3. (a) Find the tangent line to the curve $r(t) = e^{-2t} \hat{i} + \cos t \hat{j} + 3 \sin t \hat{k}$ at the point $(1, 1, 0)$ and intersects the $yz$ -plane. | 05    |
| (b) Prove that the principal normal to the helix is the normal to the cylinder.                                                                                 | 05    |
| 4. (a) Describe the smooth parametrization process of a curve.                                                                                                  | 05    |
| (b) Find the arc length of the curve $x = \cos t, y = \sin t, z = t$ from $t = 0$ to $t = \pi$ .                                                                | 05    |

### Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                  |    |
|------------------------------------------------------------------------------------------------------------------|----|
| 5. (a) State and prove Serret-Frenet formula.                                                                    | 05 |
| (b) Find the minimum value of the radius of the curvature for the curve $x = e^t, y = e^{-t}, z = \sqrt{2}t$ .   | 05 |
| 6. (a) Determine the involutes and evolutes of the twisted cubic given by $x = u, y = u^2, z = u^3$ .            | 05 |
| (b) Prove that the locus of the centre of curvature is an evolute only when the curve is plane.                  | 05 |
| 7. (a) Define curvature. Find the curvature of the circular helix: $x = a \cos t, y = a \sin t, z = ct, a > 0$ . | 05 |
| (b) State and prove Weingarten equations.                                                                        | 05 |
| 8. (a) Discuss the physical meaning of Jacobians.                                                                | 04 |
| (b) Find the radii of curvature and torsion of the helix                                                         | 06 |
| $x = a \cos u, y = a \sin u, z = au \tan \alpha$ .                                                               |    |

\*\*\*\*\* THE END \*\*\*\*\*

18/20/1

**Khulna University, Khulna**  
**Mathematics Discipline**  
 4<sup>th</sup> Year, Term-I Examination, Session: 2020-2021  
 Course No: **Math-4107**  
 Full Title of Course: **Differential Geometry**  
 Full Marks: 60, Time: 03 Hours

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

**Marks**

1. (a) What do you mean by differential geometry? Describe the branches of differential geometry. 02
- (b) Define space curves. Find the parametric equation of space curve. 03
- (c) Find the length of the curve given as the intersection of two surfaces  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ ,  $z = \cosh\left(\frac{z}{a}\right)$  from the point  $(a, 0, 0)$  to the point  $(x, y, z)$ . 05
2. (a) Give the definition of tangent line. Show that  $\frac{d\vec{r}}{ds}$  is the unit tangent vector to a space curve at the point  $p(\vec{r})$ . 05
- (b) Show that the tangent at the point of a curve of intersection of the ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  and the confocal whose parameter is  $\lambda$  can be given by  $\frac{x(X-x)}{a^2(b^2-c^2)(a^2-\lambda)} = \frac{y(Y-y)}{b^2(c^2-a^2)(b^2-\lambda)} = \frac{z(Z-z)}{c^2(a^2-b^2)(c^2-\lambda)}$ . 05
3. (a) Define curvature  $k$  and torsion  $\tau$  of a twisted curve and establish Serret-Frenet formula. 05
- (b) Prove that the osculating plane at  $(x_1, y_1, z_1)$  on the curve of intersection of the cylinders  $x^2 + z^2 = a^2$ ,  $y^2 + z^2 = b^2$  is given by  $\frac{xx_1^3 - zz_1^3 - a^4}{a^2} = \frac{yy_1^3 - zz_1^3 - b^4}{b^2}$ . 05
4. (a) Define principle normal and bi-normal of space curve. Find the equation and the direction of the principle normal and binormal of space curve. 04
- (b) Define a circular helix. Show that if the curvature and torsion are both constant then the curve is a circular helix. 04
- (c) Describe the graph of the vector valued function  $x = 1-t$ ,  $y = 3t$ ,  $z = 2t$ . 02

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Show that the necessary and sufficient condition for a curve to be a helix is that the ratios of the curvature and torsion is constant. 05
- (b) Define Bertnard curves. Explain its properties. 05
6. (a) State the fundamental theorem of space curve and prove its existence part. 05
- (b) Describe the geometrical interpretation of the second fundamental form. 05
7. (a) Define involute and evolute of space curve. Find the curvature and torsion for involute. 05
- (b) What are the spherical indicatrix? Show that the radii of curvature and torsion of the spherical indicatrix of the binormal are  $\rho_1 \sqrt{\rho^2 + \sigma^2} : \frac{(\rho^2 + \sigma^2)}{\{\sigma(\sigma\rho' - \sigma'\rho)\}}$  where  $\sigma$  and  $\rho$  have their usual meaning. 05
8. (a) Find angle between space curves. Also show that the curve bisecting the angles between the parametric curves are given by  $Edu^2 - Gdv^2 = 0$ . 05
- (b) Show that the tangent plane to the surface  $xyz = a^3$  and the coordinate planes bound a tetrahedron of constant volume. 05

**Khulna University, Khulna**  
**Mathematics Discipline**  
 4<sup>th</sup> Year, Term-II Examination, Session: 2020-2021  
 Course No: Math-4107  
 Full Title of Course: **Differential Geometry**  
 Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                                                                                                                                                                                            | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. (a) What are the functions of class- $r$ ? Deduce the formula for the arc length of a curve between two points in vector form.                                                                                                                                                          | 06    |
| (b) Find the length of the curve given as the intersection of the surfaces $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ , $x = \cosh\left(\frac{z}{a}\right)$ from the point $(a, 0, 0)$ to the point $(x, y, z)$ .                                                                             | 04    |
| 2. (a) What is a tangent line? Show that $\frac{d\vec{r}}{ds}$ is a unit tangent vector to a space curve at point $p(\vec{r})$ .                                                                                                                                                           | 05    |
| (b) Show that the tangent at a point of the curve of intersection of an ellipsoid and the confocal whose parameter is $\lambda$ can be given by<br>$\frac{x(X-x)}{a^2(b^2-c^2)(a^2-\lambda)} = \frac{y(Y-y)}{b^2(c^2-a^2)(b^2-\lambda)} = \frac{z(Z-z)}{c^2(a^2-b^2)(c^2-\lambda)}$        | 05    |
| 3. (a) Define curvature, torsion and screw curvature. Establish the serret-frenet formula at a point of a space curve.                                                                                                                                                                     | 05    |
| (b) Determine $a, h, b$ so that the paraboloid $2z = ax^2 + 2hxy + by^2$ may have the closest possible contact at the origin with the curves $x = t^3 - 2t^2 + 1, y = t^3 - 1, z = t^2 - 2t + 1$ . Also find the order of contact.                                                         | 05    |
| 4. (a) Show that $\rho = -\sigma = \frac{3}{2}(1+2u^2)^2$ for the curve $x = 3u, y = 3u^2, z = 2u^3$                                                                                                                                                                                       | 05    |
| (b) On the bi-normal of a curve of constant torsion $\tau$ , a point Q is taken at a constant distance C from the curve. Show that the bi-normal to the locus of Q is inclined to the bi-normal of the given curve at an angle $\tan^{-1}\left[\frac{c\tau^2}{k\sqrt{c^2\tau^2+1}}\right]$ | 05    |

**Section-B**

There are **FOUR** questions in these sections. Answer any **THREE** questions

- |                                                                                                                                                            |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. (a) What is intrinsic equation? State and prove fundamental theorem for space curves.                                                                   | 04 |
| (b) Define involute and evolute. Derive the equation of evolute. Also, find the curvature and torsion of evolute.                                          | 06 |
| 6. (a) Define the spherical indicatrix of the principal normal. Deduce the curvature and torsion of the spherical indicatrix of principal normal.          | 05 |
| (b) What is circular helix? Find the involutes and evolutes of the circular helix $x = a \cos \theta, y = a \sin \theta, z = a\theta \tan \alpha$ .        | 05 |
| 7. (a) Describe Monge's form of surface. Calculate the fundamental magnitudes for the Monge's form of the surface $z = f(x, y)$                            | 05 |
| (b) What do you mean by a circular helix? Obtain the equations of a circular helix $r = (a \cos u, a \sin u, bu)$ in the interval $-\infty < u < \infty$ . | 05 |
| 8. (a) Describe the geometrical interpretation of the second fundamental form.                                                                             | 05 |
| (b) Show that if L, M, N vanishes everywhere on a surface, then the surface is a part of a plane.                                                          | 05 |

\*\*\* THE END \*\*\*

**Khulna University, Khulna**  
**Mathematics Discipline**  
 4<sup>th</sup> Year, Term-II Examination, Session: 2020-2021  
 Course No: Math-4107  
 Full Title of Course: **Differential Geometry**  
 Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                                                                                                                                                                                            | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. (a) What are the functions of class- $r$ ? Deduce the formula for the arc length of a curve between two points in vector form.                                                                                                                                                          | 06    |
| (b) Find the length of the curve given as the intersection of the surfaces $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, x = \cosh\left(\frac{z}{a}\right)$ from the point $(a, 0, 0)$ to the point $(x, y, z)$ .                                                                                | 04    |
| 2. (a) What is a tangent line? Show that $\frac{d\vec{r}}{ds}$ is a unit tangent vector to a space curve at point $p(\vec{r})$ .                                                                                                                                                           | 05    |
| (b) Show that the tangent at a point of the curve of intersection of an ellipsoid and the confocal whose parameter is $\lambda$ can be given by<br>$\frac{x(X-x)}{a^2(b^2-c^2)(a^2-\lambda)} = \frac{y(Y-y)}{b^2(c^2-a^2)(b^2-\lambda)} = \frac{z(Z-z)}{c^2(a^2-b^2)(c^2-\lambda)}$        | 05    |
| 3. (a) Define curvature, torsion and screw curvature. Establish the serret-frenet formula at a point of a space curve.                                                                                                                                                                     | 05    |
| (b) Determine $a, h, b$ so that the paraboloid $2z = ax^2 + 2hxy + by^2$ may have the closest possible contact at the origin with the curves $x = t^3 - 2t^2 + 1, y = t^3 - 1, z = t^3 - 2t + 1$ . Also find the order of contact.                                                         | 05    |
| 4. (a) Show that $\rho = -\sigma = \frac{3}{2}(1+2u^2)^2$ for the curve $x = 2u, y = 3u^2, z = 2u^3$                                                                                                                                                                                       | 05    |
| (b) On the bi-normal of a curve of constant torsion $\tau$ , a point Q is taken at a constant distance C from the curve. Show that the bi-normal to the locus of Q is inclined to the bi-normal of the given curve at an angle $\tan^{-1}\left[\frac{c\tau^2}{k\sqrt{c^2\tau^2+1}}\right]$ | 05    |

**Section-B**

There are **FOUR** questions in these sections. Answer any **THREE** questions

- |                                                                                                                                                            |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. (a) What is intrinsic equation? State and prove fundamental theorem for space curves.                                                                   | 04 |
| (b) Define involute and evolute. Derive the equation of evolute. Also, find the curvature and torsion of evolute.                                          | 06 |
| 6. (a) Define the spherical indicatrix of the principal normal. Deduce the curvature and torsion of the spherical indicatrix of principal normal.          | 05 |
| (b) What is circular helix? Find the involutes and evolutes of the circular helix $x = a \cos \theta, y = a \sin \theta, z = a \theta \tan \alpha$ .       | 05 |
| 7. (a) Describe Monge's form of surface. Calculate the fundamental magnitudes for the Monge's form of the surface $z = f(x, y)$                            | 05 |
| (b) What do you mean by a circular helix? Obtain the equations of a circular helix $r = (a \cos u, a \sin u, bu)$ in the interval $-\infty < u < \infty$ . | 05 |
| 8. (a) Describe the geometrical interpretation of the second fundamental form.                                                                             | 05 |
| (b) Show that if L, M, N vanishes everywhere on a surface, then the surface is a part of a plane.                                                          | 05 |

\*\*\* THE END \*\*\*

# Khulna University, Khulna

## Mathematics Discipline

4<sup>th</sup> Year, Special Term Examination; Session: 2020-2021

Course No: Math-4107

Course Title: Differential Geometry

Full Marks: 60

Time: 03 hours

1. The figures in the margin indicate full marks. The questions are of equal value.
2. Use separate sheet for each section.

### Section A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- Q1. (a) Define space curves. Find the parametric equation of space curve. 03  
(b) Deduce the formula for the arc length of a curve between two points in vector form. 04  
(c) Determine the length of one complete turn of the circular helix:  
 $r = (a \cos u, a \sin u, cu), -\infty < u < \infty$  03
- Q2. (a) Define tangent line. Obtain the unit tangent vector to a space curve at the point  $p(\vec{r})$ . 05  
(b) Obtain the direction cosines of the tangent to the curve of intersection of the surfaces  
 $F_1(x, y, z) = 0, F_2(x, y, z) = 0$  05
- Q3. (a) Find the equation of the plane that has three point contact at the origin with the curve  
 $x = t^4 - 1, y = t^3 - 1, z = t^2 - 1$ . 05  
(b) Find the equation of osculating plane at a point  $u = 1$  of the curve  $\vec{r} = (au, -bu^2, cu^3)$  05
- Q4. (a) On the bi-normal of a curve of constant torsion  $\tau$ , a point Q is taken at a constant distance  $c$  from the curve. Show that the bi-normal to the locus of Q is inclined to the bi-normal of the given curve at an angle  $\tan^{-1} \left( \frac{c\tau^2}{\kappa\sqrt{c^2\tau^2+1}} \right)$  05  
(b) Show that the tangent at any point of the curve whose equations are  $x = 3u, y = 3u^2, z = 2u^3$  makes a constant angle with the line  $y = z - x = 0$ . 05

### Section B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- Q5. (a) Describe Monge's form of a surface. Find the equation of tangent and normal of the surface. 06  
(b) If a curve lies on a sphere show that  $\rho$  and  $\sigma$  are related by  $\frac{d}{ds}(\sigma\rho') + \frac{\rho}{\sigma} = 0$ , then show that a necessary and sufficient condition that a curve lies on a sphere is that  $\frac{d}{ds}(\frac{\rho'}{\tau}) + \frac{\rho}{\sigma} = 0$  at every point on the curve. 04
- Q6. (a) Define helices. Show that for all helices curvature bears a constant ratio with torsion. 05  
(b) Find the involutes and evolutes of the circular helix  $x = a \cos \theta, y = a \sin \theta, z = a \theta \tan \alpha$ . 05
- Q7. (a) Define spherical indicatrix. Find out the curvature and torsion of the spherical indicatrix with respect to the binormal. 05  
(b) Describe the two important properties of the first fundamental form of metric. 05
- Q8. (a) Show that a curve is a general helix if the tangent indicatrix is a circle. 05  
(b) Find the first and second fundamental magnitudes for the surface  $2z = ax^2 + 2hxy + by^2$ , where  $x, y$  are parameters. 05

\*\*\*\* End \*\*\*\*



**Khulna University, Khulna**  
**Mathematics Discipline**  
**4<sup>th</sup> Year, 2<sup>nd</sup> Term, Examination; Session: 2021-2022**  
**Course No: Math-4107, Full Title of Course: Differential Geometry**  
**Full Marks: 60, Time: 03 Hours**

- Figures in the margin indicate full marks. Questions are of equal value.
- Use separate sheet for each section.
- All the symbols have their usual meanings.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

**Marks**

1. (a) Define space curve and explain your definition by means of examples. 02  
(b) Define functions of class-r. Deduce the formula for the arc length of a curve between two points in vector form. 05  
(c) Show that  $\frac{d\vec{r}}{ds}$  is unit tangent vector to a space curve at the point  $p(\vec{r})$ . 03
2. (a) Define Osculating plane at a point on a space curve. Show that at the point  $u$  of the curve  $\underline{x} = \underline{x}(u)$  the equation to the Osculating plane is given by  $[(\underline{y} - \underline{x}) \underline{\dot{x}\dot{x}}] = 0$ . 05  
(b) The tangent at any point of the curve whose equations are  $x = 3u$ ,  $y = 3u^2$ ,  $z = 2u^3$  makes a constant angle with the line  $y = z - x = 0$ , prove it. 05
3. (a) Show that principal normal at consecutive points do not intersect unless  $\tau = 0$ . 05  
(b) Prove that for any curve  $[t', t'', t'''] = [r'', r''', r'''''] = k^3 (k\tau' - k'\tau) = k^5 \frac{d}{ds} \left( \frac{k}{\tau} \right)$  and  $[b', b'', b'''] = \tau^3 (k'\tau - k\tau') = \tau^5 \frac{d}{ds} \left( \frac{k}{\tau} \right)$ . 05
4. (a) Establish the Serret-Frenet formula at a point of a space curve. 05  
(b) Prove that necessary and sufficient condition for a curve to be a plane curve is  $[\vec{r}, \vec{r}', \vec{r}''] = 0$ . 05

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Define circular helix. Show that the equation of a circular helix is  $r = (a \cos \theta, a \sin \theta, b\theta)$ . 05  
(b) What is an intrinsic equation? State the fundamental theorem for space curves and prove its uniqueness part. 05
6. (a) Show that a necessary and sufficient condition for a curve to be helix is that the ratio of the curvature and the torsion is constant 05  
(b) What are the spherical indicatrices? Show that the radii of curvature and torsion of the spherical indicatrix of the binormal are  $\rho_1 \sqrt{(\rho^2 + \sigma^2)} : (\rho^2 + \sigma^2) / \{\sigma(\sigma\rho' - \sigma'\rho)\}$ ,  $\rho$  and  $\sigma$  having their usual meaning. 05
7. (a) Derive the equation of evolute. Also evaluate the curvature and torsion of evolute. 05  
(b) Define tangent plane and normal. Estimate the equations of the tangent plane and normal to the surface  $z = xy$  at  $(3, 5, 6)$ . 05
8. (a) Describe the two important properties of the first fundamental form of metric. 05  
(b) Calculate the fundamental magnitudes for the surface of revolution  $x = u \cos v$ ,  $y = u \sin v$ ,  $z = f(u)$  with  $u, v$  as parameters. 05

\*\*\*\*\* THE END \*\*\*\*\*



**Khulna University, Khulna**  
**Mathematics Discipline**  
**4<sup>th</sup> Year, 2<sup>nd</sup> Term, Examination; Session: 2021-2022**  
**Course No: Math-4107, Full Title of Course: Differential Geometry**  
**Full Marks: 60, Time: 03 Hours**

- Figures in the margin indicate full marks. Questions are of equal value.
- Use separate sheet for each section.
- All the symbols have their usual meanings.

**Section-A**

- There are **FOUR** questions in this section. Answer any **THREE** questions
- |                                                                                                                                                                                                                                                                    | <b>Marks</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1. (a) Define space curve and explain your definition by means of examples.                                                                                                                                                                                        | 02           |
| (b) Define functions of class-r. Deduce the formula for the arc length of a curve between two points in vector form.                                                                                                                                               | 05           |
| (c) Show that $\frac{d\vec{r}}{ds}$ is unit tangent vector to a space curve at the point $p(\vec{r})$ .                                                                                                                                                            | 03           |
| 2. (a) Define Osculating plane at a point on a space curve. Show that at the point $u$ of the curve $\underline{x} = \underline{x}(u)$ the equation to the Osculating plane is given by $[(\underline{y} - \underline{x}) \cdot \underline{\dot{x}\dot{x}}] = 0$ . | 05           |
| (b) The tangent at any point of the curve whose equations are $x = 3u$ , $y = 3u^2$ , $z = 2u^3$ makes a constant angle with the line $y = z - x = 0$ , prove it.                                                                                                  | 05           |
| 3. (a) Show that principal normal at consecutive points do not intersect unless $\tau = 0$ .                                                                                                                                                                       | 05           |
| (b) Prove that for any curve $[t', t'', t'''] = [r'', r''', r'''''] = k^3 (k\tau' - k'\tau) = k^5 \frac{d}{ds} \left( \frac{k}{\tau} \right)$ and $[b', b'', b'''] = \tau^3 (k'\tau - k\tau') = \tau^5 \frac{d}{ds} \left( \frac{k}{\tau} \right)$ .               | 05           |
| 4. (a) Establish the Serret-Frenet formula at a point of a space curve.                                                                                                                                                                                            | 05           |
| (b) Prove that necessary and sufficient condition for a curve to be a plane curve is $[\dot{r}, \ddot{r}, \ddot{\ddot{r}}] = 0$ .                                                                                                                                  | 05           |

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                                                                                                                                                                                           |    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. (a) Define circular helix. Show that the equation of a circular helix is $r = (a \cos \theta, a \sin \theta, b\theta)$ .                                                                                                                                                               | 05 |
| (b) What is an intrinsic equation? State the fundamental theorem for space curves and prove its uniqueness part.                                                                                                                                                                          | 05 |
| 6. (a) Show that a necessary and sufficient condition for a curve to be helix is that the ratio of the curvature and the torsion is constant                                                                                                                                              | 05 |
| (b) What are the spherical indicatrices? Show that the radii of curvature and torsion of the spherical indicatrix of the binormal are $\rho_1 \sqrt{(\rho^2 + \sigma^2)} : (\rho^2 + \sigma^2) / \{\sigma(\sigma\rho' - \sigma'\rho)\}$ , $\rho$ and $\sigma$ having their usual meaning. | 05 |
| 7. (a) Derive the equation of evolute. Also evaluate the curvature and torsion of evolute.                                                                                                                                                                                                | 05 |
| (b) Define tangent plane and normal. Estimate the equations of the tangent plane and normal to the surface $z = xy$ at $(3, 5, 6)$ .                                                                                                                                                      | 05 |
| 8. (a) Describe the two important properties of the first fundamental form of metric.                                                                                                                                                                                                     | 05 |
| (b) Calculate the fundamental magnitudes for the surface of revolution $x = u \cos v$ , $y = u \sin v$ , $z = f(u)$ with $u, v$ as parameters.                                                                                                                                            | 05 |

\*\*\*\*\* THE END \*\*\*\*\*

**Khulna University, Khulna**  
**Mathematics Discipline**  
**4<sup>th</sup> Year, Term-I Examination, Session: 2021-2022**  
**Course No: Math-4107, Full Title of Course: Differential Geometry**  
**Full Marks: 60, Time: 03 Hours**

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |    |                                                                                                                                                                                                                     | Marks |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. | (a) What is the differential geometry? Describe the local and the global differential geometry with examples.                                                                                                       | 04    |
|    | (b) Define space curves. Find the parametric equation of space curve.                                                                                                                                               | 03    |
|    | (c) Find the length of the curve given as the intersection of the surfaces $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $x = \cosh\left(\frac{z}{a}\right)$ from the point $(a, 0, 0)$ to the point $(x, y, z)$ .    | 03    |
| 2. | (a) Find the equation of a tangent line to a curve at a given point. Also obtain the direction cosines of the tangent to the curve of intersection of the surface $F_1(x, y, z) = 0$ and $F_2(x, y, z) = 0$ .       | 06    |
|    | (b) The tangent at any point of the curve whose equations are $x = 3u$ , $y = 3u^2$ , $z = 2u^3$ makes a constant angle with the line $y = z - x = 0$ , prove it.                                                   | 04    |
| 3. | (a) What are the three fundamental planes in differential geometry? Describe them.                                                                                                                                  | 03    |
|    | (b) Show that principal normal at consecutive points do not intersect unless $\tau = 0$ .                                                                                                                           | 03    |
|    | (c) Prove that there are three points on the cubic $x = at^3 + b$ , $y = 3ct^2 + 3dt$ , $z = 3et + f$ such that the osculating planes pass through the origin and that the points lie in the plane $3cex + afy = 0$ | 04    |
| 4. | (a) Establish the Serret-Frenet formula at a point of a space curve.                                                                                                                                                | 06    |
|    | (b) Prove that necessary and sufficient condition for a curve to be a plane curve is $[\dot{r}, \ddot{r}, \ddot{\ddot{r}}] = 0$ .                                                                                   | 04    |

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |    |                                                                                                                                                                |    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. | (a) What is a circular helix? Deduce the equations of a circular helix.                                                                                        | 03 |
|    | (b) Show that for all helices curvature bears a constant ratio with torsion                                                                                    | 03 |
|    | (c) Prove that the curve $x = au$ , $y = bu^2$ , $z = cu^3$ is a helix iff $3ac = \pm 2b^2$ .                                                                  | 04 |
| 6. | (a) If two curves have same intrinsic equations then they are congruent, prove it.                                                                             | 05 |
|    | (b) When a curve lies on a sphere, then prove that $\rho$ and $\sigma$ are related by $\frac{d}{ds}(\sigma\rho) + \frac{\rho}{\sigma} = 0$ .                   | 05 |
| 7. | (a) Derive the equation of evolute. Also find the curvature and torsion of evolute.                                                                            | 05 |
|    | (b) Define tangent plane and normal. Find the equations of the tangent plane and normal to the surface $z = xy$ at $(2, 3, 6)$ .                               | 05 |
| 8. | (a) Describe Monge's form of surface. Show that the tangent plane to the surface $xyz = a^3$ and the coordinate planes bound a tetrahedron of constant volume. | 05 |
|    | (b) Calculate the fundamental magnitudes and the normal of the surface $2z = ax^2 + 2hxy + by^2$                                                               | 05 |

END



**Khulna University, Khulna**  
**Mathematics Discipline**  
 4<sup>th</sup> Year, Term-I Examination 2024; Session: 202-2023  
 Course No: Math-4107  
 Full Title of Course: Differential Geometry  
 Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |    |                                                                                                                                                                                                                                                                                       | Marks |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. | (a) Define a space curve and explain your definition by means of examples.                                                                                                                                                                                                            | 03    |
|    | (b) Deduce the formula for the arc length of a curve between two points in vector form.                                                                                                                                                                                               | 04    |
|    | (c) Determine the length of one complete turn of the circular helix,<br>$r = (a \cos u, a \sin u, cu), -\infty < u < \infty$ .                                                                                                                                                        | 03    |
| 2. | (a) Define the osculating plane at a point of space curve and determine its equation.                                                                                                                                                                                                 | 03    |
|    | (b) Show that when the curve is analytic, there exist a definite osculating plane at a point of inflexion $P$ provided that the curve is not a straight line.                                                                                                                         | 03    |
|    | (c) Determine $a, h, b$ so that the paraboloid $2z = ax^2 + 2hxy + by^2$ may have the closest possible contact at the origin with the curves $x = t^3 - 2t^2 + 1, y = t^3 - 1, z = t^2 - 2t + 1$ . Also find the order of contact.                                                    | 04    |
| 3. | (a) Define principle normal and bi-normal of space curve. Find the equation and the direction of the principle normal and bi-normal of space curve.                                                                                                                                   | 04    |
|    | (b) Prove that there are three points on the cubic $x = at^3 + b, y = 3ct^2 + 3dt, z = 3et + f$ such that the osculating planes pass through the origin and that the points lie in the plane $3cex + afy = 0$ .                                                                       | 06    |
| 4. | (a) Find the radii of curvature and torsion at any point of the curve $x^2 + y^2 = a^2, x^2 - y^2 = az$ .                                                                                                                                                                             | 05    |
|    | (b) Given the curve $r = (e^{-u} \sin u, e^{-u} \cos u, e^{-u})$ . Find at any point $u$ of this curve (i) the equation of the tangent, (ii) the equation of the normal plane, (iii) the curvature, (iv) the unit principle normal vector $n$ , (v) the equation of Principle vector. | 05    |

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |    |                                                                                                                                                                                                                                                                                                                                                          |    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. | (a) If a curve is drawn on any cylinder and makes a constant angle $\alpha$ with the generators, show that $\rho = \rho_0 \cos^2 \alpha$ and $\sigma = \rho_0 \operatorname{cosec} \alpha \sec \alpha$ , where $\frac{1}{\rho}$ and $\frac{1}{\rho_0}$ are curvatures of any point $P$ of the curve and the normal section of the cylinder through $P$ . | 05 |
|    | (b) Obtain the equation of a circular helix $r = (a \cos u, a \sin u, bu), -\infty < u < \infty$ where $a > 0$ referred to $s$ as parameter and show that the length of one complete turn of the helix is $2\pi c$ where $c = \sqrt{a^2 + b^2}$ .                                                                                                        | 05 |
| 6. | (a) Find the equation for the tangent plane to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$ .                                                                                                                                                                                                                                                   | 05 |
|    | (b) Find a unit normal vector to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$ .                                                                                                                                                                                                                                                          | 05 |
| 7. | (a) Prove that if $\kappa(s)$ and $\tau(s)$ are continuous functions of a real variable $s (s \geq 0)$ , then there exists a space curve for which $\kappa$ is the curvature, $\tau$ is the torsion, and $s$ is the arc length measured from some suitable base point.                                                                                   | 05 |
|    | (b) Derive the equation of evolute. Find the curvature and torsion of evolute.                                                                                                                                                                                                                                                                           | 05 |
| 8. | (a) Define Bertnard Curves with graph. Examine $x = a \cos t, y = \sin t, z = ct$ is a plane curve or not.                                                                                                                                                                                                                                               | 05 |
|    | (b) Show that the curve bisecting the angles between the parametric curves are given by $E du^2 - G dv^2 = 0$ .                                                                                                                                                                                                                                          | 05 |

\*\*\*\*\* THE END \*\*\*\*\*



**Khulna University, Khulna**  
**Mathematics Discipline**  
 4<sup>th</sup> Year, Term-I Examination 2024; Session: 202<sup>2</sup>-2023  
 Course No: Math-4107  
 Full Title of Course: Differential Geometry  
 Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                                                                                                                                                                                       | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. (a) Define a space curve and explain your definition by means of examples.                                                                                                                                                                                                         | 03    |
| (b) Deduce the formula for the arc length of a curve between two points in vector form.                                                                                                                                                                                               | 04    |
| (c) Determine the length of one complete turn of the circular helix,<br>$r = (a \cos u, a \sin u, cu), -\infty < u < \infty$ .                                                                                                                                                        | 03    |
| 2. (a) Define the osculating plane at a point of space curve and determine its equation.                                                                                                                                                                                              | 03    |
| (b) Show that when the curve is analytic, there exist a definite osculating plane at a point of inflexion $P$ provided that the curve is not a straight line.                                                                                                                         | 03    |
| (c) Determine $a, h, b$ so that the paraboloid $2z = ax^2 + 2hxy + by^2$ may have the closest possible contact at the origin with the curves $x = t^3 - 2t^2 + 1, y = t^3 - 1, z = t^2 - 2t + 1$ . Also find the order of contact.                                                    | 04    |
| 3. (a) Define principle normal and bi-normal of space curve. Find the equation and the direction of the principle normal and bi-normal of space curve.                                                                                                                                | 04    |
| (b) Prove that there are three points on the cubic $x = at^3 + b, y = 3ct^2 + 3dt, z = 3et + f$ such that the osculating planes pass through the origin and that the points lie in the plane $3cex + afy = 0$ .                                                                       | 06    |
| 4. (a) Find the radii of curvature and torsion at any point of the curve $x^2 + y^2 = a^2, x^2 - y^2 = az$ .                                                                                                                                                                          | 05    |
| (b) Given the curve $r = (e^{-u} \sin u, e^{-u} \cos u, e^{-u})$ . Find at any point $u$ of this curve (i) the equation of the tangent, (ii) the equation of the normal plane, (iii) the curvature, (iv) the unit principle normal vector $n$ , (v) the equation of Principle vector. | 05    |

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |                                                                                                                                                                                                                                                                                                                                                                             |    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. (a) If a curve is drawn on any cylinder and makes a constant angle $\alpha$ with the generators, show that $\rho = \rho_0 \operatorname{cosec}^2 \alpha$ and $\sigma = \rho_0 \operatorname{cosec} \alpha \sec \alpha$ , where $\frac{1}{\rho}$ and $\frac{1}{\rho_0}$ are curvatures of any point $P$ of the curve and the normal section of the cylinder through $P$ . | 05 |
| (b) Obtain the equation of a circular helix $r = (a \cos u, a \sin u, bu), -\infty < u < \infty$ where $a > 0$ referred to $s$ as parameter and show that the length of one complete turn of the helix is $2\pi c$ where $c = \sqrt{a^2 + b^2}$ .                                                                                                                           | 05 |
| 6. (a) Find the equation for the tangent plane to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$ .                                                                                                                                                                                                                                                                   | 05 |
| (b) Find a unit normal vector to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$ .                                                                                                                                                                                                                                                                             | 05 |
| 7. (a) Prove that if $\kappa(s)$ and $\tau(s)$ are continuous functions of a real variable $s (s \geq 0)$ , then there exists a space curve for which $\kappa$ is the curvature, $\tau$ is the torsion, and $s$ is the arc length measured from some suitable base point.                                                                                                   | 05 |
| (b) Derive the equation of evolute. Find the curvature and torsion of evolute.                                                                                                                                                                                                                                                                                              | 05 |
| 8. (a) Define Bertnard Curves with graph. Examine $x = a \cos t, y = \sin t, z = ct$ is a plane curve or not.                                                                                                                                                                                                                                                               | 05 |
| (b) Show that the curve bisecting the angles between the parametric curves are given by $E du^2 - G dv^2 = 0$ .                                                                                                                                                                                                                                                             | 05 |

\*\*\*\*\* THE END \*\*\*\*\*

Khulna University, Khulna  
Mathematics Discipline  
4<sup>th</sup> Year, Term-I, Examination Session: 2023-2024  
Course No: Math-4107  
Full Title of Course: Differential Geometry  
Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

|                                                                                                                                                                                                                                                                                                        | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. (a) Describe differential geometry with its branches.                                                                                                                                                                                                                                               | 02    |
| (b) Deduce the formula for the arc length of a curve between two points in vector form.                                                                                                                                                                                                                | 05    |
| (c) Find the length of the circular helix<br>$r(u) = a \cos u \mathbf{i} + a \sin u \mathbf{j} + cu \mathbf{k}$ , $-\infty < u < \infty$ from $(a, 0, 0)$ to $(a, 0, 2\pi c)$ .<br>Also obtain its equation in terms of parameter $s$ .                                                                | 03    |
| 2. (a) Show that if a curve is given in terms of a general parameter $u$ , then equation of the osculating plane is $[R - r, \dot{r}, \ddot{r}] = 0$ .                                                                                                                                                 | 03    |
| (b) Define contact of $n^{\text{th}}$ order of a curve and a surface. Also find the conditions for this type of contact.                                                                                                                                                                               | 03    |
| (c) Find the lines that have four point contact at $(0, 0, 1)$ with the surface<br>$x^4 + 3xyz + x^2 - y^2 - z^2 + 2yz - 3xy - 2y + 2z = 1$                                                                                                                                                            | 04    |
| 3. (a) Describe the unit vectors $\mathbf{t}, \mathbf{n}, \mathbf{b}$ and fundamental planes.                                                                                                                                                                                                          | 03    |
| (b) Determine the direction of the principal normal and the bi-normal.                                                                                                                                                                                                                                 | 02    |
| (c) State and explain the Serret-Frenet formula.                                                                                                                                                                                                                                                       | 05    |
| 4. (a) For a point on the curve of intersection of the surfaces $x^2 - y^2 = c^2, y = x \tanh(z/c)$<br>prove that $\rho = -\sigma = 2x^2/c$ .                                                                                                                                                          | 05    |
| (b) On the bi-normal of a curve of constant torsion $\tau$ , a point Q is taken at a constant distance $c$ from the curve. Show that the bi-normal to the locus of Q is inclined to the bi-normal of the given curve at an angle $\tan^{-1} \left[ \frac{c\tau^2}{\kappa\sqrt{c^2\tau^2+1}} \right]$ . | 05    |

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

|                                                                                                                                                                                                                                                                                              |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. (a) Estimate the equations of a circular helix. A curve is drawn on a parabolic cylinder to cut all the generators at the same angle. Find expression for the curvature and torsion.                                                                                                      | 05 |
| (b) Show that the curvature of the helix $x = a \cos \theta, y = a \sin \theta, z = a\theta \tan \alpha$ is $\frac{\cos^2 \alpha}{a}$ , also estimate the torsion.                                                                                                                           | 05 |
| 6. (a) State and prove the fundamental theorem of space curves.                                                                                                                                                                                                                              | 05 |
| (b) What are the spherical indicatrices? Show that the radii of curvature and torsion of the spherical indicatrix of the binormal are<br>$\rho_1 \sqrt{(\rho^2 + \sigma^2)} : (\rho^2 + \sigma^2) / \{\sigma(\sigma\rho' - \sigma'\rho)\}$ , $\rho$ and $\sigma$ having their usual meaning. | 05 |
| 7. (a) Determine the point $P(1, 0, 0)$ of the position vector $\vec{r}(u, v) = 2u\mathbf{i} + 3(u+v)\mathbf{j} + uv\mathbf{k}$ , $r(u, v) = 2u\mathbf{i} + 3(u+v)\mathbf{j} + uv\mathbf{k}$ either regular or singular point.                                                               | 05 |
| (b) Find the equation of the osculating sphere and osculating circle at $(1, 2, 3)$ on the curve<br>$x = 2t + 1, y = 3t^2 + 2, z = 4t^2 + 3$ .                                                                                                                                               | 05 |
| 8. (a) If a geodesic on a surface of revolution cuts the meridians at a constant angle, then prove that the surface is a right cylinder.                                                                                                                                                     | 05 |
| (b) The necessary and sufficient condition for a curve on a sphere to be geodesic is that the curve is a great circle, prove it.                                                                                                                                                             | 05 |

\*\*\*\*\* THE END \*\*\*\*\*

**Khulna University**  
 Science, Engineering and Technology School  
 Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**  
 Year: **Fourth**, Term: **I** Examination-2026; Session: **2024-2025**  
 Course Code: **0541 12 Math 4111**  
 Course Title: **Differential Geometry**  
 Total Marks: **60**, Time: **03** Hours, Credit: **03**



- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

- |                                                                                                                                                                                                     |    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1. (a) Define functions of class-r and equivalent paths. [L1; CLO1]                                                                                                                                 | 02 |
| (b) Deduce the formula for the arc length of a curve between two points in vector form. [L4; CLO1]                                                                                                  | 04 |
| (c) Find the arc length of the curve given as the intersection of the surfaces $x^2 + y^2 = a^2$ , $z = \tan^{-1}\left(\frac{y}{x}\right)$ , from the point $(a, 0, 0)$ to $(x, y, z)$ . [L1; CLO1] | 04 |
| 2. (a) Define tangent line. [L1; CLO2]                                                                                                                                                              | 01 |
| (b) Show that $\frac{d\vec{r}}{ds}$ is unit tangent vector to a space curve at the point $p(\vec{r})$ . [L4; CLO2]                                                                                  | 04 |
| (c) Obtain the direction cosines of the tangent to the curve of intersection of the surfaces $F_1(x, y, z) = 0$ , $F_2(x, y, z) = 0$ . [L5; CLO2]                                                   | 05 |
| 3. (a) Define principle normal and bi-normal of space curve. [L1; CLO3]                                                                                                                             | 02 |
| (b) Find the equation and the direction of the principle normal and bi-normal of space curve. [L1; CLO3]                                                                                            | 04 |
| (c) Find the inflexional tangents at $(x_1, y_1, z_1)$ on the surface $y^2z = 4ax$ . [L1; CLO3]                                                                                                     | 04 |
| 4. (a) Define the osculating plane of a curve at a point. Find its equation in terms of $\vec{r}'(t)$ and $\vec{r}''(t)$ . When does the osculating plane not exist? [L1; CLO2]                     | 03 |
| (b) Show that if a curve lies in a plane, its binormal vector $\vec{B}(t)$ is constant. Conversely, if $\vec{B}(t)$ is constant, prove the curve is planar. [L2; CLO2]                              | 03 |
| (c) Derive the Frenet-Serret formulas. [L6; CLO2]                                                                                                                                                   | 04 |

### Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

**Marks**

- |    |     |                                                                                                                                                                                                                              |    |
|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. | (a) | Define circular helix. [L1; CLO5]                                                                                                                                                                                            | 01 |
|    | (b) | Find the equation of a circular helix. [L1; CLO5]                                                                                                                                                                            | 04 |
|    | (c) | Show that the curvature and torsion of the helix $x = a \cos \theta, y = a \sin \theta, z = a\theta \tan \alpha$ are $\frac{\cos^2 \alpha}{a}$ and $\pm \frac{\sin \alpha \cos \alpha}{a}$ , respectively. [L4; CLO5]        | 05 |
| 6. | (a) | Define intrinsic equation. [L1; CLO6]                                                                                                                                                                                        | 02 |
|    | (b) | State and prove the Uniqueness theorem for space curves. [L5; CLO6]                                                                                                                                                          | 04 |
|    | (c) | Obtain the curvature $\kappa_1$ and torsion $\tau_1$ of the involute. [L5; CLO6]                                                                                                                                             | 04 |
| 7. | (a) | Define the circle of curvature for a space curve at a point where curvature $\kappa > 0$ . Explain why the circle of curvature lies in the osculating plane and has radius $\frac{1}{\kappa}$ . Why is it unique? [L1; CLO1] | 03 |
|    | (b) | Define the spherical indicatrix of the binormal $B(s)$ . Also, deduce the curvature and torsion of the spherical indicatrix of binormal. [L1; CLO2]                                                                          | 03 |
|    | (c) | Find a unit normal vector to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$ . [L1; CLO7]                                                                                                                       | 04 |
| 8. | (a) | Explain that the first fundamental form is a quadratic form that measures arc length on the surface. Why is it called a "metric"? [L3; CLO4]                                                                                 | 03 |
|    | (b) | Compute the first fundamental form for the surface $\mathbf{r}(u, v) = (u \cos v, u \sin v, u^2)$ at a general point $(u, v)$ . [L3; CLO4]                                                                                   | 04 |
|    | (c) | Show that the curve bisecting the angles between the parametric curves are given by $E du^2 - G dv^2 = 0$ . [L4; CLO8]                                                                                                       | 03 |

\*\*\*\*\* THE END \*\*\*\*\*

**Khulna University**  
 Science, Engineering and Technology School  
 Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**  
 Year: **Fourth Term: I Examination-2026; Session: 2024-2025**  
 Course Code: **Math-4107**



Course Title: **Differential Geometry**  
 Total Marks: **60**, Time: **03 Hours**, Credit: **03**

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

**Section-A**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |    |                                                                                                                                                                                                      | <b>Marks</b> |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1. | (a) Define the contact of $n$ -th order between a curve and a surface in parametric form.                                                                                                            | 03           |
|    | (b) Show that a curve and a surface have contact of order 2 iff the curve lies in the tangent plane of the surface at the point.                                                                     | 03           |
|    | (c) Find the length of the curve given as the intersection of the surfaces $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ , $x = \cosh\left(\frac{z}{a}\right)$ from the point $(a, 0, 0)$ to $(x, y, z)$ . | 04           |
| 2. | (a) Show that if a curve is given in terms of a general parameter $u$ , then equation of the osculating plane is $[R - r, \dot{r}, \ddot{r}] = 0$                                                    | 03           |
|    | (b) Define contact of $n^{th}$ order of a curve and a surface and find the conditions for this type of contact.                                                                                      | 03           |
|    | (c) Find the lines that have four point contact at $(0,0,1)$ with the surface $x^4 + 3xyz + x^2 - y^2 - z^2 + 2yz - 3xy - 2y + 2z = 1$                                                               | 04           |
| 3. | (a) Describe the unit vectors $t, n, b$ and fundamental planes.                                                                                                                                      | 03           |
|    | (b) Determine the direction of the principal normal and the bi-normal.                                                                                                                               | 02           |
|    | (c) State and explain the Serret-Frennet's formula.                                                                                                                                                  | 05           |
| 4. | (a) Find Curvature and Torsion of a curve for both $r = f(u)$ and $r = f(s)$                                                                                                                         | 05           |
|    | (b) If $C$ is a curve for which $b$ varies differentially with arc length. Then to show that a necessary and sufficient condition that $C$ is a plane curve is that $\tau = 0$ at all points.        | 05           |

**Section-B**

There are **FOUR** questions in this section. Answer any **THREE** questions

- |    |                                                                                                                                                                                         |    |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5. | (a) Estimate the equations of a circular helix. A curve is drawn on a parabolic cylinder to cut all the generators at the same angle. Find an expression for the curvature and torsion. | 05 |
|    | (b) Show that the curvature of the helix $x = a \cos \theta, y = a \sin \theta, z = a \theta \tan \alpha$ is $\frac{\cos^2 \alpha}{a}$ , also estimate the torsion.                     | 05 |
| 6. | (a) What is intrinsic equation?                                                                                                                                                         | 02 |
|    | (b) State and prove the fundamental theorem of space curves.                                                                                                                            | 04 |
|    | (c) Write the properties of Bertrand curves.                                                                                                                                            | 04 |
| 7. | (a) Derive the equation of evolute. Also evaluate the curvature and torsion of evolute.                                                                                                 | 05 |
|    | (b) Find the center and radius of circle of curvature.                                                                                                                                  | 05 |
| 8. | (a) Define the spherical indicatrix (Spherical image) of the tangent.                                                                                                                   | 02 |
|    | (b) Deduce the curvature and torsion of the spherical indicatrix of Tangent.                                                                                                            | 04 |
|    | (c) Describe the two important properties of the first fundamental form of metric.                                                                                                      | 04 |

\*\*\*\*\* THE END \*\*\*\*\*